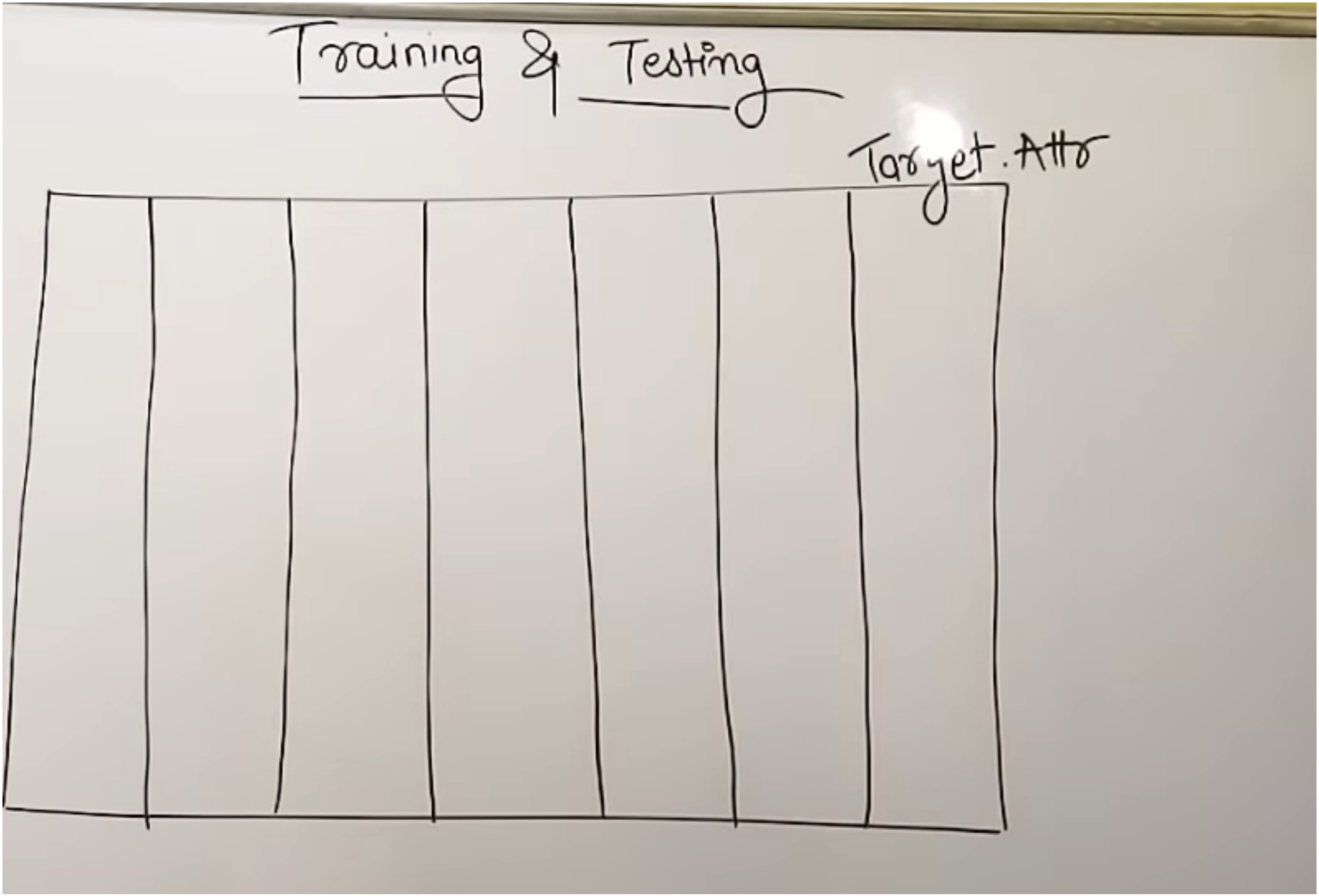
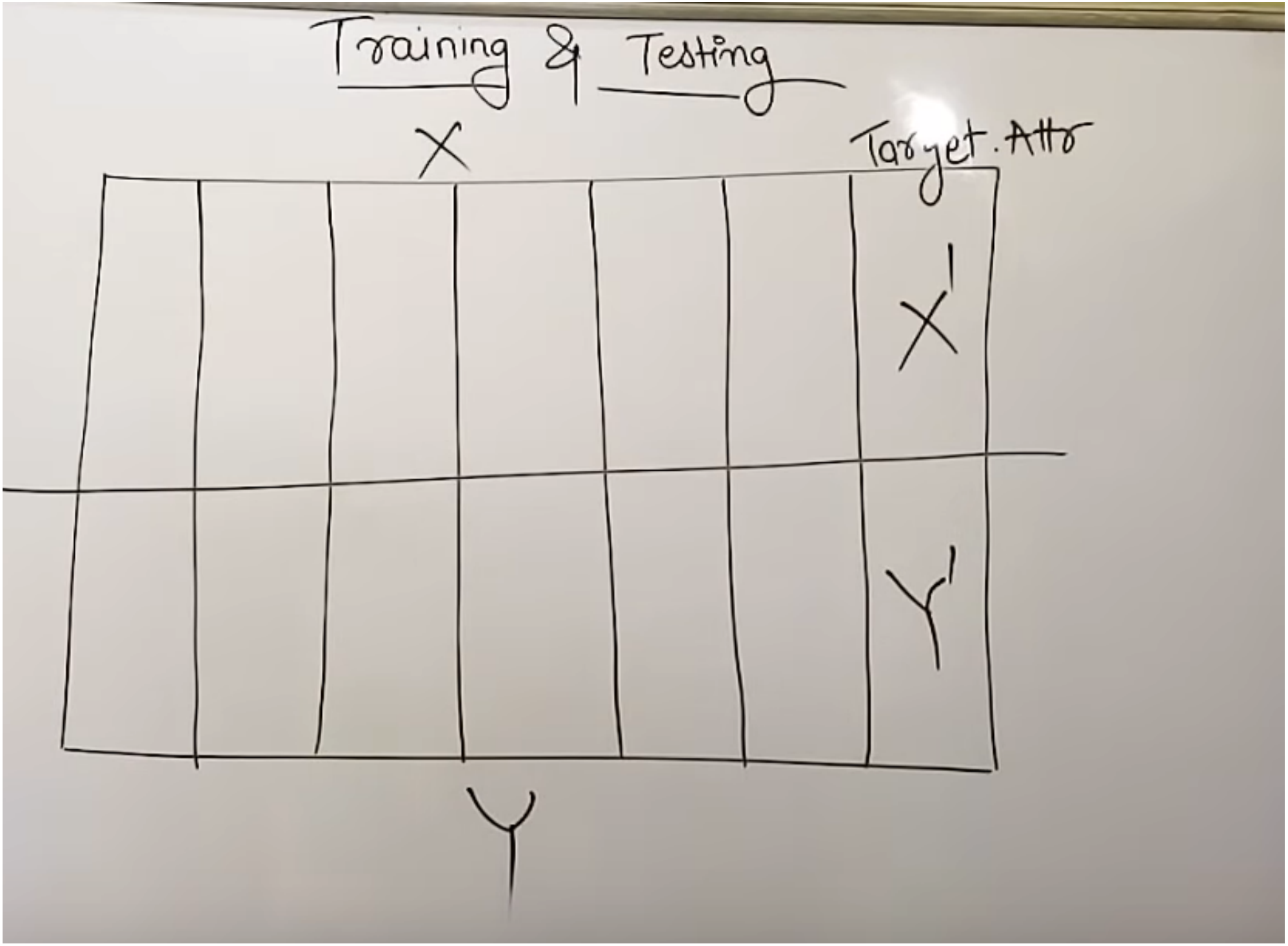
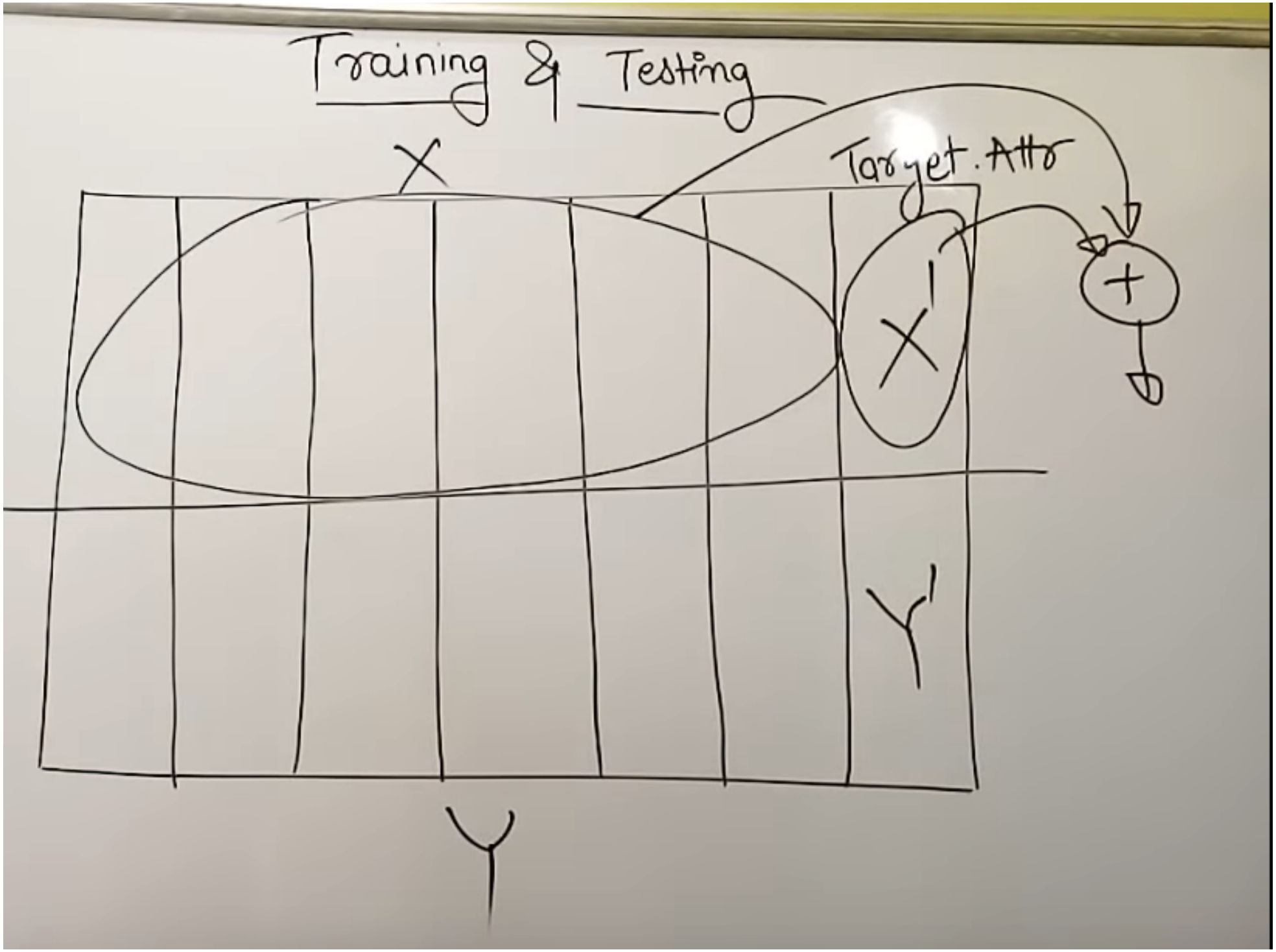
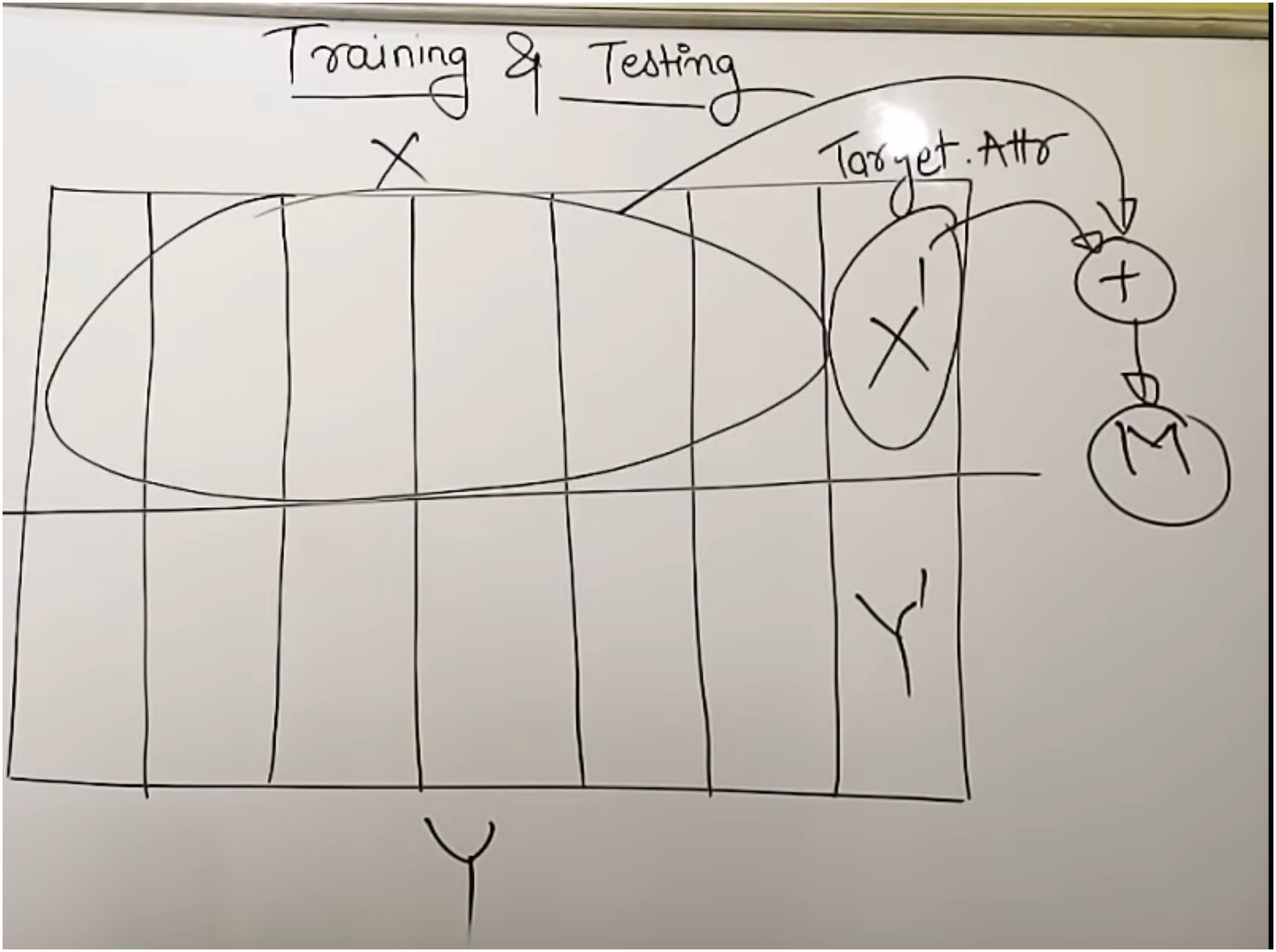
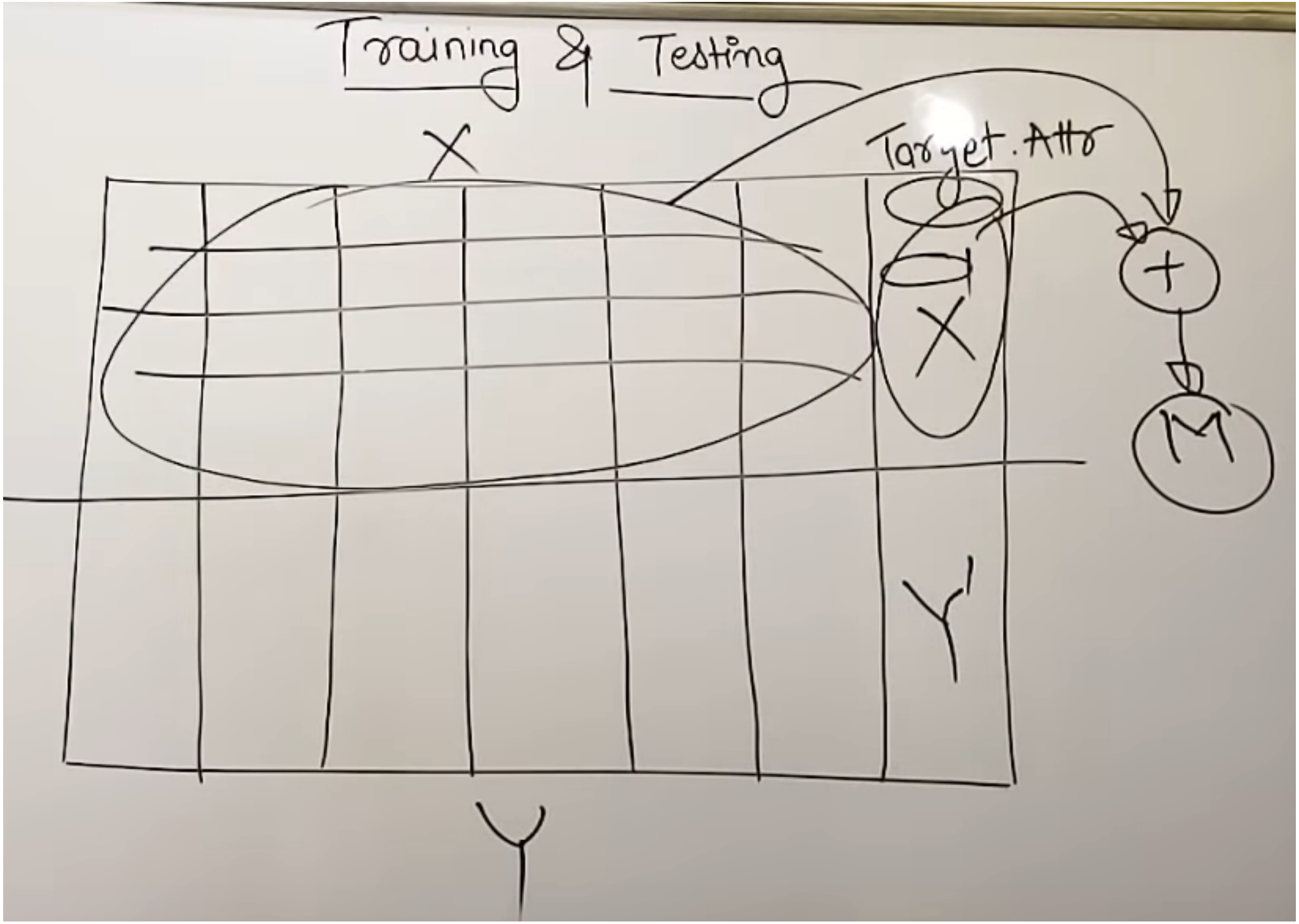
Training & Testing

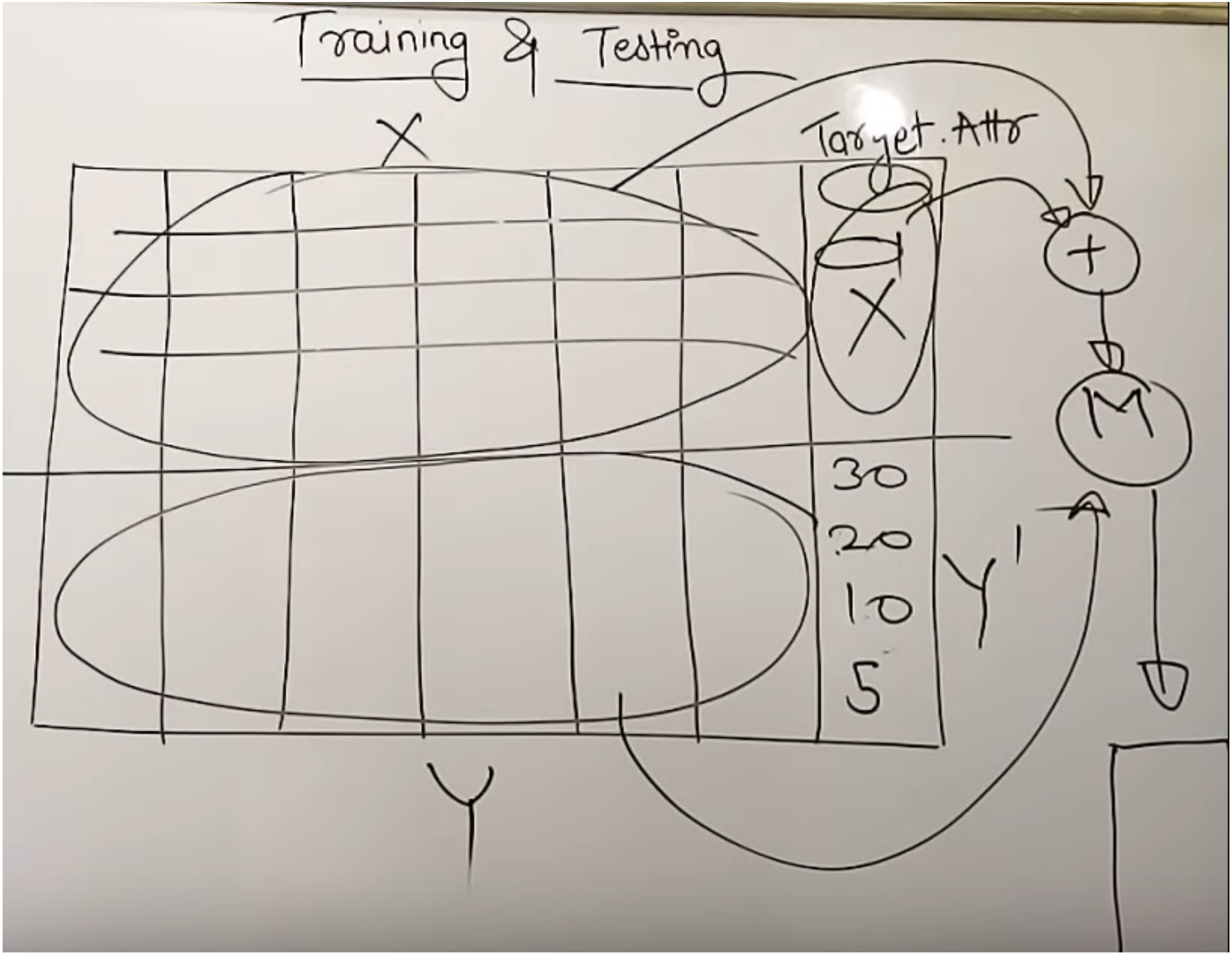


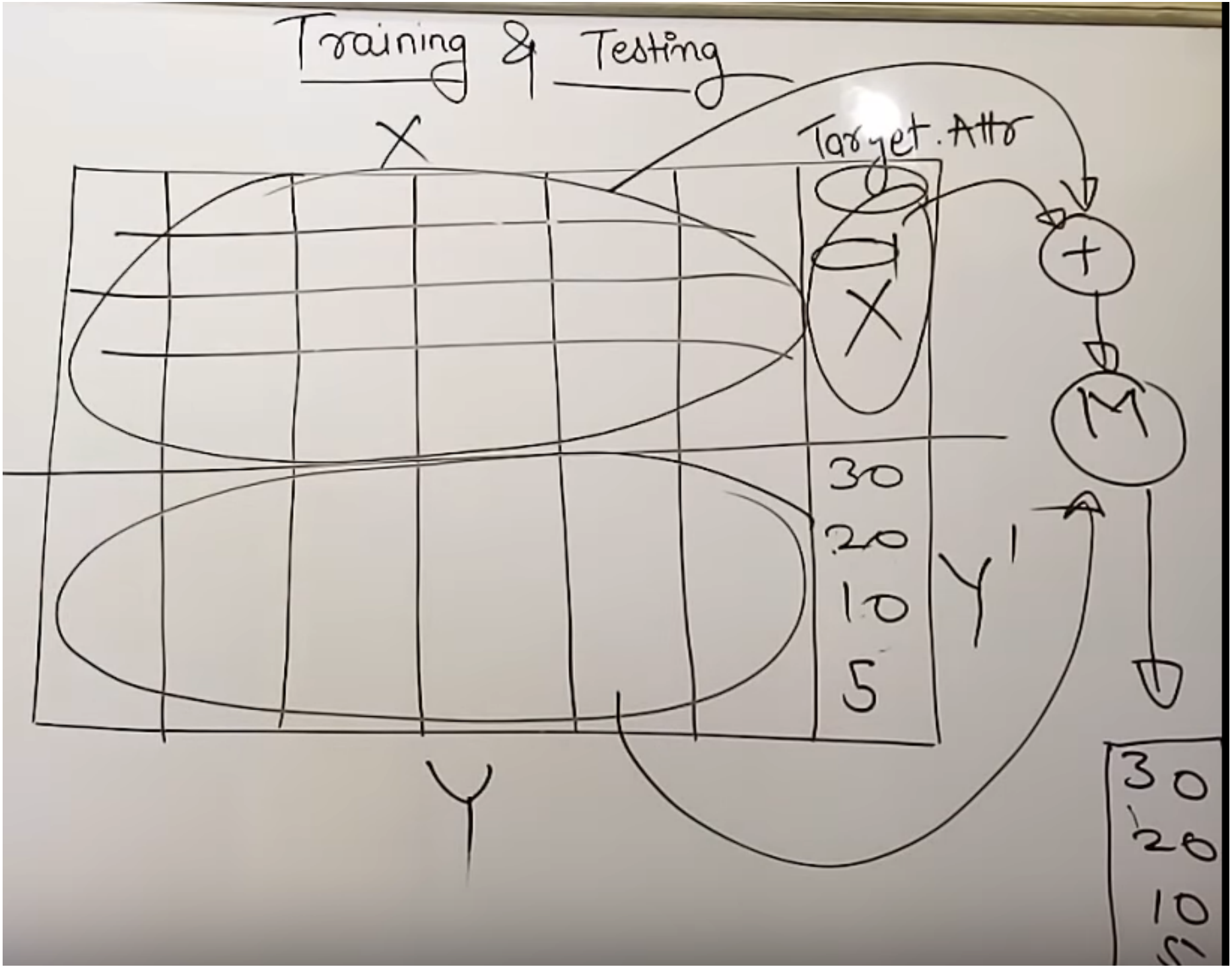

Training & Testing
X
Target. Attr
X'
Y'
Y

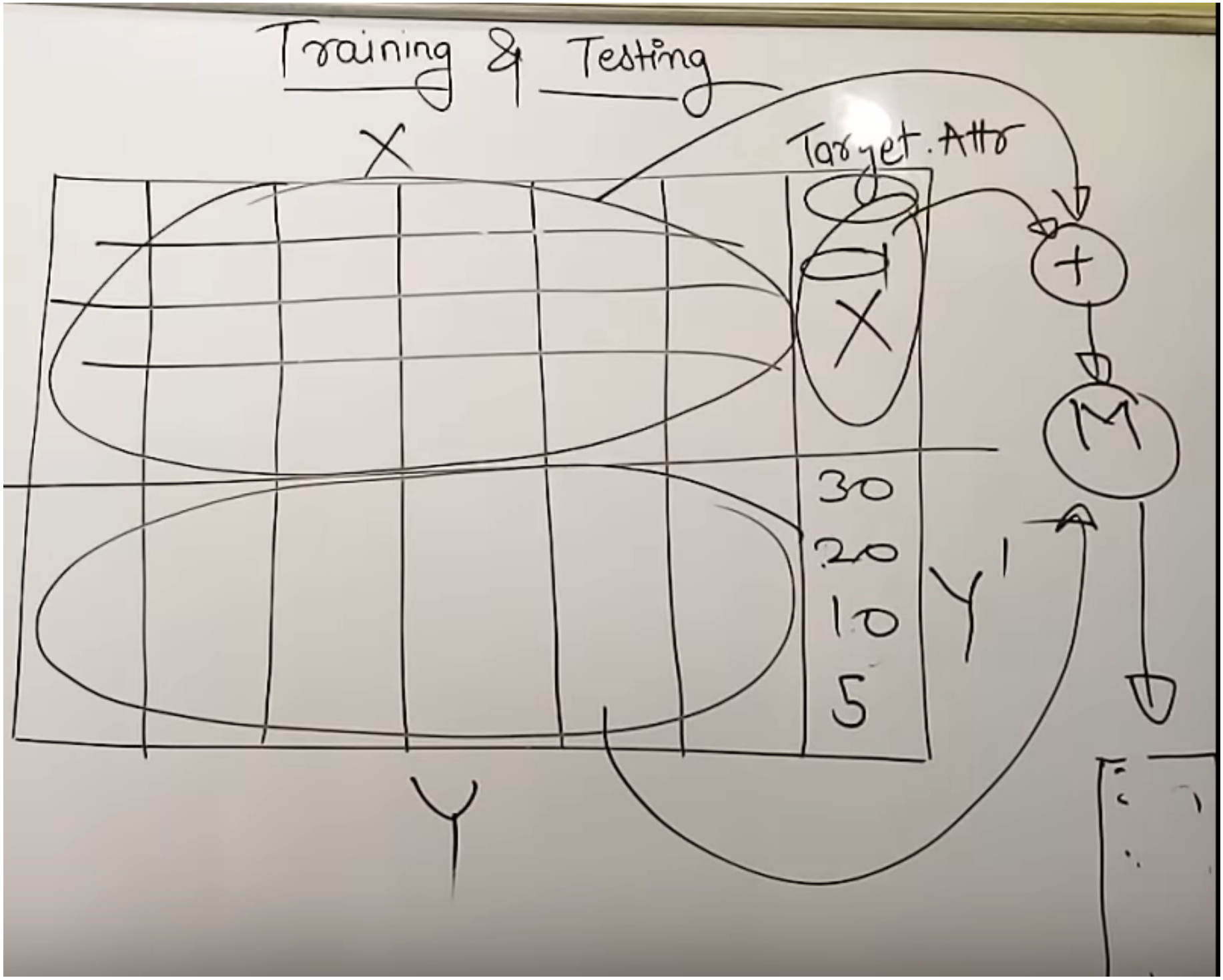

Training & Testing
X
Target Attr
X'
+
Y'
Y


Training & Testing
X
Target. Attr
X'
+
M
Y'
Y


Training & Testing
X
Target. Attr
X
+
M
Y'
Y


Training & Testing
X
Target Attr
30
20
10
5
Y
+
M
Y'


Training & Testing
X
Target. Attr
X
30
20
10
5
Y
Y
+
M
30
20
10


Training & Testing
X
Target. Attr
X
30
20
10
5
Y
Y'
+
M

training & testing data.

Features

Definition: *Features are the individual input variables or attributes that the machine learning algorithm uses to make predictions. These are the characteristics or properties of the data that the algorithm analyzes.*

Example: *In a dataset of houses, features could include square footage, number of bedrooms, number of bathrooms, and location.*

Labels (Target Variable or Output)

Definition: *Labels are the output variables that the machine learning algorithm is trying to predict. They represent the target or the outcome of interest.*

Example: *In a dataset of houses, the label could be the sale price.*

Training Data

Definition: *Training data is the portion of the dataset that is used to train or teach the machine learning algorithm. It consists of both features and corresponding labels. The algorithm learns patterns and relationships in the training data.*

Purpose: *The goal is for the algorithm to generalize from the training data and make accurate predictions on new, unseen data.*

Example: *In a dataset of houses, 80% of the data might be used for training the algorithm.*

Testing Data

Definition: *Testing data is a separate portion of the dataset that is not used during the training phase. It is reserved to evaluate the performance of the trained algorithm. Testing data includes features and corresponding labels.*

Purpose: *The testing data helps assess how well the algorithm generalizes to new, unseen data and whether it can make accurate predictions.*

Example: *In a dataset of houses, 20% of the data might be set aside for testing the algorithm.*

Workflow

Data Splitting:

- *The dataset is typically split into a training set and a testing set. The training set is used for training the model, and the testing set is used for evaluating its performance.*
- *Common split ratios are 80/20 or 70/30, depending on the size of the dataset.*

Training:

- *The algorithm is trained on the training data by learning patterns and relationships between features and labels.*

Testing/Evaluation:

- *The trained model is then tested on the separate testing data to assess its performance. This involves making predictions on the testing features and comparing them to the actual labels.*

Performance Metrics:

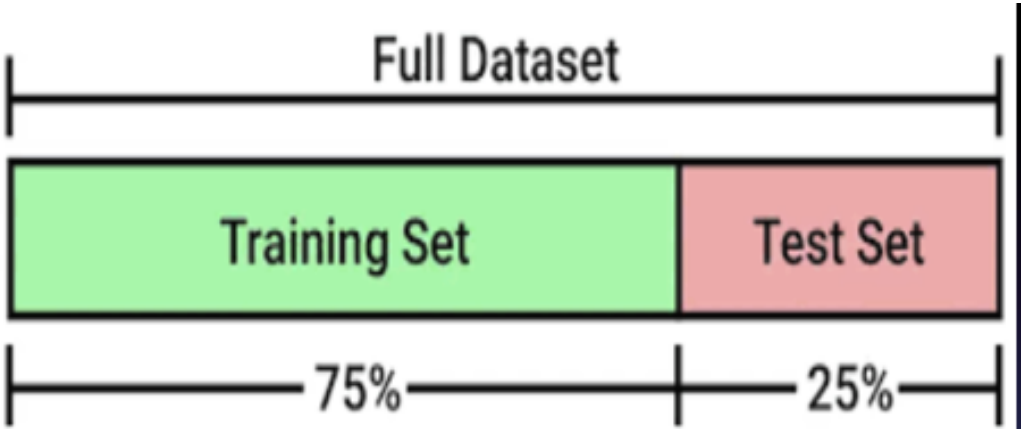
- *Metrics such as accuracy, precision, recall, and F1-score are often used to quantify the performance of the model on the testing data.*

Training a Model

Let say we have sale price, sq ft., year built, number of rooms, tax information of 100 houses.




Training a Model
Let say we have sale price, sq ft., year built, number of rooms, tax information of 100 houses.
Initial Data
=
Training Data
+
Test Data
100 houses
75 houses
25 houses


Full Dataset
Training Set
Test Set
75%
25%

Train-Test Split

Split your dataset into two parts: a training set used to train the model and a test set used to evaluate its performance.

```
from sklearn.model_selection import
train_test_split

X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2,
random_state=42)
```